

Total no. of Pages: 2

Roll no.

Degree: B-Tech.

Semester 3rd

END-SEMESTER EXAMINATION, June, 2026

Course Title: Probability Theory and Random Process

Course Code: Sem: ECECC303/EIECC303/VTECC303/ECECC06

Duration: 3.00 Hours

Max. Marks: 50

Note: - Attempt all questions in the given order only. Missing data/information (if any), may be suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO																				
1.	Attempt any two parts of the following:																						
1a	Let X be a continuous random variable with the following distribution: $f(x) = \begin{cases} kx, & \text{if } 0 \leq x \leq 5 \\ 0, & \text{elsewhere} \end{cases}$ (a) Evaluate k . (b) Find $P(1 \leq X \leq 3)$. (c) Find $P(X \leq 3)$.	5	C01																				
1b	Let X and Y be independent random variables. Then show that (i) $E[XY] = E[X] E[Y]$, (ii) $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$, (iii) $\text{Cov}(X, Y) = 0$.	5	C01																				
1c	Find the expectation (μ), variance (σ^2), and standard deviation (σ) of the following distribution <table border="1" style="margin: 10px auto;"> <tr> <td>x_i</td><td>2</td><td>3</td><td>11</td></tr> <tr> <td>$f(x_i)$</td><td>$\frac{1}{8}$</td><td>$\frac{1}{4}$</td><td>$\frac{1}{8}$</td></tr> </table>	x_i	2	3	11	$f(x_i)$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{8}$	5	C01												
x_i	2	3	11																				
$f(x_i)$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{8}$																				
2.	Attempt any two parts of the following:																						
2a	Let X and Y be random variables with joint distribution given below: <table border="1" style="margin: 10px auto;"> <tr> <td>$X \backslash Y$</td><td>-3</td><td>2</td><td>4</td><td>Sum</td></tr> <tr> <td>1</td><td>0.1</td><td>0.2</td><td>0.2</td><td>0.5</td></tr> <tr> <td>3</td><td>0.3</td><td>0.1</td><td>0.1</td><td>0.5</td></tr> <tr> <td>Sum</td><td>0.4</td><td>0.3</td><td>0.3</td><td></td></tr> </table> Find: (a) the distributions of X and Y , (b) $\text{Cov}(X, Y)$, (c) $\rho(X, Y)$	$X \backslash Y$	-3	2	4	Sum	1	0.1	0.2	0.2	0.5	3	0.3	0.1	0.1	0.5	Sum	0.4	0.3	0.3		5	C02
$X \backslash Y$	-3	2	4	Sum																			
1	0.1	0.2	0.2	0.5																			
3	0.3	0.1	0.1	0.5																			
Sum	0.4	0.3	0.3																				
2b	Team A has probability $2/3$ of winning whenever it plays. If A plays 4 games, find the probability that A wins (i) exactly 2 games, (ii) at least 1 game, (iii) more than half of the games.	5	C02																				
2c	Let X be a random variable with the Poisson distribution $p(k, \lambda)$. Then show that (i) $E(X) = \lambda$ and (ii) $\text{Var}(X) = \lambda$, (iii) $\sigma_X = \sqrt{\lambda}$	5	C02																				
3.	Attempt any two parts of the following:																						
3a	State and prove Chebyshev inequality	5	C03																				

3b	Let X be a random variable with the following distribution and let $Y = X^2$. Determine (i) the distribution (g) of Y (ii) the joint distribution (h) of X and Y , (iii) and $\rho(X, Y)$.	5	C03										
	<table><tr><td>x_i</td><td>-2</td><td>-1</td><td>1</td><td>2</td></tr><tr><td>$f(x_i)$</td><td>$\frac{1}{4}$</td><td>$\frac{1}{4}$</td><td>$\frac{1}{4}$</td><td>$\frac{1}{4}$</td></tr></table>	x_i	-2	-1	1	2	$f(x_i)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$		
x_i	-2	-1	1	2									
$f(x_i)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$									
3c	The joint PDF of the random variables X and Y is given as $f_{XY}(x, y) = \frac{1}{4}e^{- x - y }$, $-\infty < x < \infty$, $-\infty < y < \infty$. Determine (i) Whether the random variables X and Y are statistically independent, (ii) The probability that $X \leq 1$ and $Y \leq 0$.	5	C03										
4.	Attempt any two parts of the following:												
4a	Define the following terms in detail : (i) Stationary random Process (ii) Ergodic process	5	C04										
4b	A random process is expressed as $X(t) = A \cos(\omega t + \theta)$, where, ω and θ are constants and A is a random variable. Determine whether $X(t)$ is wide sensed stationary WSS process.	5	C04										
4c	If the random process is wide sense stationary than prove that $R_{XX}(-\tau) = R_{XX}(\tau)$	5	C04										
5.	Attempt any two parts of the following:												
5a	Derive the expression for power spectral density of a wide-sensed process.	5	C05										
5b	Define the autocorrelation function of a random process, derive its expression and explain its significance.	5	C05										
5c	Discuss cross correlation and Gaussian process in detail.	5	C05										